

Name: Mayienli Nuella

Country: Cameroon

Track: Data Analytics

Week 1 Requirements

A. Understand the Business Problem:

1. What problem is FinTrust trying to solve?

FinTrust is trying to solve the problem of not fully turning its customer and transaction data into timely, trustworthy, and actionable intelligence.

More specifically, it wants to:

- Understand customer behavior and transaction patterns better.
- Detect and manage risk earlier, such as fraud, money laundering, credit risk, and suspicious activity.
- Improve operational efficiency by reducing manual reviews and slow processes.
- Strengthen digital customer support through faster, more personalized, and scalable responses.
- Use ML and generative AI responsibly to support better decisions across risk, operations, and customer service.

In short: FinTrust wants to move from reactive, siloed, manual decision-making to proactive, data-driven, intelligent decision-making.

2. Why is the problem important?

This problem is important because it directly affects FinTrust's money, compliance, customers, and reputation.

- **Financial impact:** Fraud, defaults, chargebacks, and inefficient operations cause direct losses.
- **Regulatory risk:** Banks must comply with AML, KYC, data privacy, fair lending, and explainability requirements.
- **Customer trust:** Slow support, false fraud alerts, or poor personalisation can drive customers away.
- **Operational cost:** Manual reviews and repetitive support tickets are expensive and do not scale.
- **Competitive pressure:** Digital banks that use data and AI well can respond faster, personalise better, and operate cheaper.
- **AI opportunity and risk:** ML and GenAI can create value, but only if the data is clean, models are monitored, and outputs are fair, explainable, and safe.

3. How can my track contribute to solving it?

My track is data analytics, it can contribute by turning raw customer and transaction data into useful signals, models, and tools.

Key contributions include :

- **Data preparation:** Clean, merge, and validate customer and transaction data; handle missing values, duplicates, outliers, and class imbalance.
- **Exploratory analysis:** Identify customer segments, transaction trends, risk patterns, and operational bottlenecks.
- **Feature engineering:** Build features such as transaction velocity, frequency, amount deviation, merchant category, device/geolocation changes, and time-based behaviour.
- **Predictive modelling:** Build models for fraud detection, AML alerts, credit risk scoring, churn prediction, and customer lifetime value.
- **Anomaly detection:** Flag unusual transactions or behaviour for review.
- **Customer support AI:** Use NLP and GenAI for intent classification, ticket summarisation, sentiment analysis, agent assist, and chatbot responses.
- **Process optimisation:** Analyse support and operations workflows to find delays, rework, and automation opportunities.
- **Responsible AI:** Evaluate fairness, explainability, false positives, and model drift; keep humans in the loop for high-risk decisions.

4. What type of output could my track provide?

My track could provide a mix of descriptive, predictive, prescriptive, and generative outputs.

Examples include:

- **Dashboards and reports:** Customer 360 views, transaction trends, risk KPIs, support performance, and operational metrics.
- **Risk scores and alerts:** Fraud probability scores, AML risk flags, credit risk tiers, and suspicious activity alerts.
- **Customer segments:** High-value customers, at-risk customers, frequent complainers, dormant accounts, and high-risk profiles.
- **Predictive lists:** Customers likely to churn, default, accept an offer, or need support.
- **Recommendations:** Next-best-action for agents, collections, retention, or cross-sell.
- **GenAI support outputs:** Draft customer replies, ticket summaries, policy-based answers, and agent copilot suggestions.
- **Model documentation:** Model cards, explainability reports, fairness checks, and monitoring dashboards.
- **Business recommendations:** Process improvements, policy changes, and automation opportunities.

The overall output would help FinTrust make **faster, safer, more personalised, and more cost-effective decisions** while maintaining trust and regulatory compliance.

B. Review the Relevant Resources

1. Identify the resources relevant to your track and explain:

For my track- data analytics, the two primary resources are the **Customer Information** dataset and the **Transaction Information** dataset. I would also create a derived **Customer–Transaction Analytical Base Table** by joining them. Below is a review of each.

a) Customer Information Resource

What it contains

A customer-level master dataset with:

- **Customer_ID** – primary key
- **Customer_Name** – PII
- **Age** – demographic
- **Gender** – demographic
- **City** – location
- **Customer_Segment** – e.g. retail, SME, premium
- **Account_Type** – product type
- **Tenure_Months** – relationship length
- **Digital_Engagement_Score** – likely a precomputed engagement metric
- **Monthly_Income_Band** – ordinal income range
- **Preferred_Channel** – e.g. mobile, web, branch, agent
- **Account_Status** – active, dormant, closed, etc.

How I expect to use it

- Build a **Customer 360 view** for segmentation and profiling.
- Create **static features** for ML models: age band, tenure, engagement score, income band, account type, preferred channel.
- Support **risk stratification**: e.g. high-tenure vs low-tenure customers, high digital engagement vs low engagement.
- Improve **customer support**: route customers to preferred channels, personalise responses, prioritise vulnerable or high-value customers.

- Run **fairness and bias checks** across age, gender, city, and income band.
- Join to transactions using **Customer_ID** to create behavioural features.

Limitations identified

- Contains **PII and sensitive attributes** — strict access control, masking, and compliance with NDPR/GDPR-type rules are required.
- **Monthly_Income_Band** is coarse, so precise affordability or credit risk modelling is limited.
- **Digital_Engagement_Score** may be opaque or precomputed; without its definition, explainability is limited.
- No **KYC risk rating, PEP/sanctions status, credit score, or product holdings** beyond account type.
- No **customer contact history, complaints, or support tickets**, limiting GenAI support use cases.
- Likely a **current snapshot**, not historical, so customer status changes over time are missing.
- Missing values, duplicates, and inconsistent city/segment labels may exist.
- Demographic variables can introduce **bias** if used carelessly in risk models.

b) Transaction Information Resource

What it contains

A transaction-level ledger with:

- **Transaction_ID** – primary key
- **Customer_ID** – foreign key to customer data
- **Transaction_DateTime** – timestamp
- **Transaction_Type** – e.g. transfer, withdrawal, deposit, card payment
- **Amount_NGN** – transaction amount in Nigerian Naira
- **Channel** – e.g. mobile app, USSD, ATM, web, branch
- **Device_Type** – e.g. Android, iOS, web
- **Location** – transaction location
- **International_Transaction** – yes/no flag
- **Transaction_Status** – success, failed, pending, reversed

- **Risk_Review_Flag** – indicates whether the transaction was flagged for review

How I expect to use it

- Build **fraud and AML anomaly detection models** using amount, velocity, frequency, channel, device, location, and international flag.
- Engineer **behavioural features**: transaction count per day/week/month, average amount, amount deviation from customer baseline, international ratio, device diversity, location changes, night-time activity, failed transaction rate, risk review rate.
- Detect **unusual patterns** such as sudden high-value transfers, rapid successive transactions, or new device/location combinations.
- Monitor **operational performance**: success/failure rates by channel, device, and location.
- Prioritise **manual risk reviews** to reduce false positives and analyst workload.
- Provide **GenAI context**: summarise a customer's recent transaction behaviour for support agents or investigators.

Limitations identified

- **Risk_Review_Flag is not a confirmed fraud label** — it may reflect manual review, bias, or incomplete investigation. Using it as a target can create label bias.
- **Class imbalance** is likely: fraudulent/suspicious transactions are rare.
- No **merchant category, counterparty, beneficiary, or narrative**, so it is hard to know *where* money went or *why*.
- No **account balance after transaction**, so cannot assess affordability, overdraft, or utilisation.
- **Location** may be city-level or inconsistent, limiting geospatial precision.
- **Device_Type** is not a unique device ID, so device fingerprinting is limited.
- **Transaction_Status** does not explain *why* a transaction failed.
- **International_Transaction** may be derived or missing for some records.
- Time zone and timestamp quality issues may affect velocity calculations.
- Amounts are only in **NGN**, so multi-currency behaviour is invisible.
- Joining transaction and customer data increases **re-identification and privacy risk**.

c) Derived Resource: Customer–Transaction Analytical Base Table

What it contains

A joined analytical table using **Customer_ID**, combining customer attributes with transaction history and engineered features such as:

- Recency, frequency, monetary value (RFM)
- Transaction velocity and amount deviation
- International transaction ratio
- Channel and device diversity
- Location change frequency
- Risk review rate
- Failed transaction rate
- Digital engagement vs actual transaction behaviour

How I expect to use it

- Train ML models for **fraud, AML, credit risk, churn, and next-best-action**.
- Power **dashboards** for risk, operations, and customer support.
- Feed **GenAI copilots** with structured customer context for summarisation, agent assist, and personalised responses.
- Support **prescriptive analytics**: which customers to review, retain, or contact.

Limitations identified

- **Join integrity** issues: missing Customer_IDs, duplicates, orphan transactions.
- **Temporal leakage**: using future transactions to predict past events.
- **Aggregation bias**: customer-level features may hide transaction-level anomalies.
- **PII concentration**: joining datasets creates a richer but more sensitive resource.
- Requires strong **data governance, feature versioning, and monitoring**.

d) Supplementary Resources Needed

To fully solve FinTrust's problems, these additional resources would be valuable but are **not provided**:

- Confirmed fraud/AML case outcomes and investigation notes

- Customer support tickets, chat logs, call transcripts
- Merchant/counterparty and beneficiary data
- Device fingerprint, IP, and geolocation history
- Credit bureau, KYC, PEP, and sanctions data
- Policy, regulatory, and product documentation for GenAI grounding

Without them, models can still provide useful signals, but **label quality, explainability, and support automation** will be limited.

Summary

The **Customer Information** resource explains **who the customer is**.

The **Transaction Information** resource explains **what the customer does**.

Together, they enable customer segmentation, fraud/AML detection, risk scoring, operational monitoring, and GenAI-assisted support. The main limitations are **PII sensitivity, coarse income/location data, missing merchant/counterparty detail, imbalanced and potentially biased risk labels, and the need for supplementary outcome and support data**.

C. Define Your Track Objectives

1. Build a governed, analysis-ready Customer–Transaction dataset

Join customer and transaction resources into a clean analytical base table with documented data quality, feature definitions, and privacy controls.

2. Develop and evaluate risk-focused ML models

Create baseline and machine learning models for transaction anomaly/fraud/AML detection and customer risk segmentation, with explainability and performance benchmarks against simple rules.

3. Generate actionable customer and operational insights

Identify customer segments, transaction behaviour patterns, channel/device performance, and drivers of risk review flags to support faster risk and operational decisions.

4. Prototype GenAI-assisted customer support and investigator tools

Build a safe prototype for transaction summarisation, support intent classification, and agent response drafting, with human-in-the-loop review.

5. Establish responsible AI and monitoring foundations

Define fairness, explainability, drift, PII protection, and feedback mechanisms so models and GenAI outputs remain compliant, trustworthy, and useful.

D. Identify Success Criteria

I will judge success across five dimensions:

Dimension	Success Criteria
Data quality	Join rate ≥ 95%, duplicate Customer_ID/Transaction_ID < 1%, critical missing fields < 5%, documented feature dictionary
Model performance	Better than baseline rules on PR-AUC/recall at fixed precision; e.g. recall ≥ 80% at precision ≥ 50% for high-risk alerts; calibrated and explainable outputs
Business impact	Reduction in manual review workload by 15–30%, faster risk triage, improved detection of suspicious patterns, reduced support handling time
Customer/agent experience	Agents and investigators find outputs useful; GenAI summaries are accurate, grounded, and require limited correction; improved response consistency
Responsible AI & compliance	No PII leakage, fairness gaps monitored across age/gender/city, human-in-the-loop for high-risk decisions, audit logs, model cards, and drift monitoring in place

E. Develop an Initial Plan

Building on Week 1’s problem framing and resource review, the plan for Weeks 2–4 is:

Week 2 — Data Foundation and Exploratory Analysis

Focus: Understand, clean, and prepare the data.

Activities:

- Audit both datasets for missing values, duplicates, inconsistent labels, and PII exposure.
- Clean and standardise fields: dates, city, channel, device, transaction type, status, risk flag.
- Join Customer and Transaction data using `Customer_ID` to create an analytical base table.
- **Perform EDA:**
 - Transaction volume, value, and frequency by customer segment.
 - Risk review rate by channel, device, location, international flag.
 - Customer engagement vs transaction behaviour.
 - Failed/pending transaction patterns.
- **Engineer first feature set:**
 - RFM (Recency, Frequency, Monetary).
 - Transaction velocity, amount deviation from customer baseline.
 - International transaction ratio, device/channel diversity, location change frequency.
 - Risk review rate and failed transaction rate.
- Define baseline rules and KPIs.

Deliverables:

- Data quality report.
- Cleaned and joined analytical base table.
- Feature dictionary.
- EDA notebook/dashboard.
- Baseline risk and operational metrics.

Week 3 — Modelling and GenAI Prototyping

Focus: Build models, segments, and GenAI proof-of-concept.

Activities:

- **Train baseline models:**

- Logistic Regression for interpretability.
- Random Forest / XGBoost for performance.
- Isolation Forest for anomaly detection.
- **Evaluate using:**
 - PR-AUC, recall at fixed precision, F1, calibration.
 - SHAP/feature importance for explainability.
- **Build customer segmentation using RFM and clustering (e.g. K-Means).**
- **Analyse operational hotspots:**
 - High-failure channels/devices.
 - Manual review concentration.
- **Prototype GenAI support tool:**
 - Transaction summarisation for agents.
 - Support intent classification.
 - Draft response generation with policy grounding.
- **Define GenAI evaluation rubric: accuracy, groundedness, hallucination rate, tone, compliance.**

Deliverables:

- Model results and comparison report.
- Customer segmentation output.
- GenAI prototype and evaluation rubric.
- Operational insight summary.

Week 4 — Validation, Integration, and Recommendations

Focus: Validate, package, and present the solution.

Activities:

- **Validate best models on holdout/time-based split.**
- **Run fairness and bias checks across age, gender, city, and income band.**

- **Stress-test GenAI outputs for hallucination, PII leakage, and unsafe responses.**
- **Build a dashboard/scorecard for:**
 - Risk alerts and model scores.
 - Customer segments and behaviour.
 - Operational performance.
 - Support/GenAI metrics.
- **Define monitoring plan:**
 - Drift detection, performance tracking, feedback loop, retraining triggers.
- **Prepare final recommendations and implementation roadmap.**
- **Document limitations, assumptions, and next steps.**

Deliverables:

- Final validated model(s) and model cards.
- Risk/operations/support dashboard.
- GenAI guardrails and monitoring plan.
- Final presentation with business recommendations.
- Implementation roadmap.

4. DATA ANALYTICS TRACK

Part A — Business Understanding

1. What business questions should FinTrust's management be able to answer?

Management should be able to answer questions across customer intelligence, transaction behaviour, risk, operations, and customer support.

Customer intelligence

- Who are our customers by segment, age, gender, city, income band, tenure, and account type?
- Which customer segments are most engaged, most valuable, or most at risk of becoming dormant?
- How does digital engagement vary by segment, channel preference, and account status?
- Which customers prefer mobile, web, branch, or agent channels?

- How does tenure affect transaction activity, risk, and retention?

Transaction behaviour

- What are the transaction volumes, values, and frequencies by customer segment, transaction type, channel, device, and location?
- Which channels and devices are most used for high-value or international transactions?
- What is the success, failure, pending, and reversal rate by channel, device, and location?
- How do customer transaction patterns change over time?

Risk and compliance

- Which transactions and customers are most likely to be flagged for risk review?
- What patterns are associated with suspicious, fraudulent, or high-risk activity?
- Which customer profiles, channels, devices, locations, or transaction types carry the highest risk?
- Are risk review flags concentrated among certain demographic groups or segments?
- What is the operational workload created by manual risk reviews?

Operations and customer support

- Which channels, devices, or locations create the most failed or problematic transactions?
- Where are the bottlenecks in transaction processing or risk review?
- Which customers need support most often, and through which channels?
- Can support queries be classified, prioritised, or automated using GenAI?
- How can agents get faster, accurate summaries of customer and transaction history?

Growth, retention, and product strategy

- Which segments should be targeted for cross-sell, up-sell, or retention?
- Which account types and channels drive the strongest engagement?
- Which customers are likely to churn, default, or become inactive?
- What products or channel improvements would best serve high-value or underserved segments?

2. What decisions could data analysis support?

Data analysis could support both strategic and operational decisions.

Decision Area	Example Decisions Supported
Customer segmentation	Which segments to target, retain, or prioritise for service
Risk management	Which transactions/customers to review first; how to tune fraud/AML rules
Fraud & AML	Which alerts to escalate; which patterns to monitor; where to deploy investigators
Customer support	How to route tickets, staff channels, and deploy chatbots/GenAI assistants
Channel strategy	Where to invest in mobile, web, agent, or branch support
Operations	How to reduce failed transactions and manual review bottlenecks
Product strategy	Which account types or features to promote, redesign, or retire
Retention & growth	Which customers to target with retention offers or cross-sell campaigns
Compliance & fairness	Whether risk decisions show bias across age, gender, city, or income band
AI/automation	Where to automate safely and where to keep humans in the loop

3. What stakeholders could benefit from the analytical outputs?

Stakeholder	How They Benefit
Executive leadership / Board	Clear KPIs on customers, risk, operations, and growth for strategic decisions
Risk & Compliance	Better monitoring of fraud, AML, regulatory exposure, and fairness
Fraud / AML analysts	Prioritised alerts, risk scores, and transaction summaries

Stakeholder	How They Benefit
Operations teams	Insight into failed transactions, bottlenecks, and process efficiency
Customer support / Contact centre	Customer context, intent classification, and GenAI-assisted responses
Product & Digital teams	Understanding of channel/device usage and feature adoption
Marketing & Customer Insights	Segmentation, targeting, retention, and campaign optimisation
Data / AI / IT teams	Clean data, feature definitions, model monitoring, and governance
Finance	Visibility into transaction value, loss risk, and cost of manual reviews
Auditors / Regulators	Evidence of controls, explainability, and responsible AI practices
Customers	Faster support, fewer false declines, better digital experiences, and fair treatment

In summary,

The Data Analytics Track should turn FinTrust’s customer and transaction data into business intelligence that helps management understand customers, manage risk, improve operations, strengthen support, and make more informed strategic decisions.

Part C — Analytical Questions

The following 8 analytical questions are designed to be answerable using the supplied **Customer Information** and **Transaction Information** datasets. They focus on descriptive, comparative, and diagnostic analysis rather than causal claims.

	Analytical Question	Area Covered	Key Fields Used
1	How do customer segments differ in transaction frequency, average transaction value, and preferred channel/device usage, and	Customer behaviour	Customer_Segment, Digital_Engagement_Score, Preferred_Channel, Transaction_DateTime,

	Analytical Question	Area Covered	Key Fields Used
	does Digital_Engagement_Score align with actual transaction activity?		Amount_NGN, Channel, Device_Type
2	What are the trends, peaks, and changes in transaction volume, value, and type over time by customer segment, account type, and channel?	Transaction activity	Transaction_DateTime, Transaction_Type, Amount_NGN, Customer_Segment, Account_Type, Channel
3	Which customer segments, account types, transaction types, channels, and locations account for the highest share of total and high-value transaction value, and how concentrated is transaction value?	Transaction value	Amount_NGN, Customer_Segment, Account_Type, Transaction_Type, Channel, Location
4	Which channels and device types have the highest transaction success rates, failure/pending/reversal rates, and average transaction amounts, and how does this vary by location?	Transaction channels & status	Channel, Device_Type, Transaction_Status, Amount_NGN, Location
5	How do international transactions differ from domestic transactions in average amount, channel/device usage, transaction status, and risk review rate?	Transaction value, channels, risk	International_Transaction, Amount_NGN, Channel, Device_Type, Transaction_Status, Risk_Review_Flag
6	What customer attributes and transaction characteristics are most strongly associated with Risk_Review_Flag ?	Risk-related patterns	Age, Gender, City, Customer_Segment, Account_Type, Tenure_Months, Digital_Engagement_Score, Monthly_Income_Band, Preferred_Channel, Account_Status, Transaction_Type, Amount_NGN, Channel, Device_Type, Location,

	Analytical Question	Area Covered	Key Fields Used
			International_Transaction, Transaction_Status
7	How do transaction status outcomes relate to risk review flags, transaction amount, channel, device type, and international status?	Transaction status & risk	Transaction_Status, Risk_Review_Flag, Amount_NGN, Channel, Device_Type, International_Transaction
8	How do customer tenure and account status relate to transaction frequency, transaction value, digital engagement, and risk review rates?	Customer behaviour & risk	Tenure_Months, Account_Status, Digital_Engagement_Score, Transaction_DateTime, Amount_NGN, Risk_Review_Flag

Part D — KPI Planning

Below are **8 KPIs** that can be calculated from the supplied customer and transaction data. They cover customer behaviour, transaction activity, value, channels, status, and risk patterns.

KPI	Definition	Why It Matters	Required Data
1. Average Transaction Value (ATV)	Total Amount_NGN for successful transactions ÷ number of successful transactions, segmented by customer segment, channel, transaction type, and account type.	Shows where value is concentrated, which segments/channels drive higher-value activity, and where	Amount_NGN, Transaction_Status, Customer_Segment, Account_Type, Channel, Transaction_Type, Customer_ID

KPI	Definition	Why It Matters	Required Data
		risk exposure may be greater.	
2. Transaction Success Rate	(Successful transactions ÷ total transactions) × 100, overall and by Channel, Device_Type, Location, and Transaction_Type.	Measures operational reliability and customer experience. Low success rates reveal friction, failed payments, or process issues.	Transaction_Status, Transaction_ID, Channel, Device_Type, Location, Transaction_Type
3. Risk Review Rate	(Transactions with Risk_Review_Flag = Yes ÷ total transactions) × 100, overall and by segment, channel, device, location, and international status.	Quantifies risk-review workload and highlights where suspicious/high-risk	Risk_Review_Flag, Transaction_ID, Customer_ID, Customer_Segment, Channel, Device_Type, Location, International_Transaction

KPI	Definition	Why It Matters	Required Data
		activity is concentrated. Supports investigator prioritisation and rule tuning.	
4. High-Value Transaction Share	Percentage of total transaction value coming from transactions above a defined threshold (e.g., top 10% or above a NGN amount), by segment and channel.	Reveals concentration of value and risk. Helps management monitor large transactions and prioritise high-value customers.	Amount_NGN, Transaction_ID, Customer_ID, Customer_Segment, Channel, Transaction_Type
5. Customer Activity Rate	Percentage of customers with at least one transaction in a given period	Measures engagement, dormant	Customer_ID, Transaction_DateTime, Account_Status, Customer_Segment, Tenure_Months

KPI	Definition	Why It Matters	Required Data
	(e.g., 30 or 90 days), by segment, tenure, and account status.	cy, and retention risk. Useful for targeting retention, reactivation, and cross-sell campaigns.	
6. Digital Engagement Alignment Score	Correlation or ratio between Digital_Engagement_Score and actual transaction frequency/value per customer.	Tests whether the engagement score reflects real behavior. Helps identify digitally engaged but inactive customers or highly active low-engagement	Digital_Engagement_Score, Customer_ID, Transaction_DateTime, Amount_NGN, Preferred_Channel

KPI	Definition	Why It Matters	Required Data
		ment custom ers.	
7. International Transaction Risk Ratio	Risk review rate for international transactions ÷ risk review rate for domestic transactions.	Highlights cross-border risk exposure and helps focus AML/fraud controls on international activity.	International_Transaction, Risk_Review_Flag, Amount_NGN, Channel, Device_Type, Transaction_Status
8. Channel/Device Failure Rate	(Failed + Reversed + Pending transactions ÷ total transactions) × 100, by Channel and Device_Type.	Identifies operational weaknesses and customer friction points. Supports channel investment, support routing,	Transaction_Status, Channel, Device_Type, Location, Transaction_ID

KPI	Definition	Why It Matters	Required Data
		and technical fixes.	

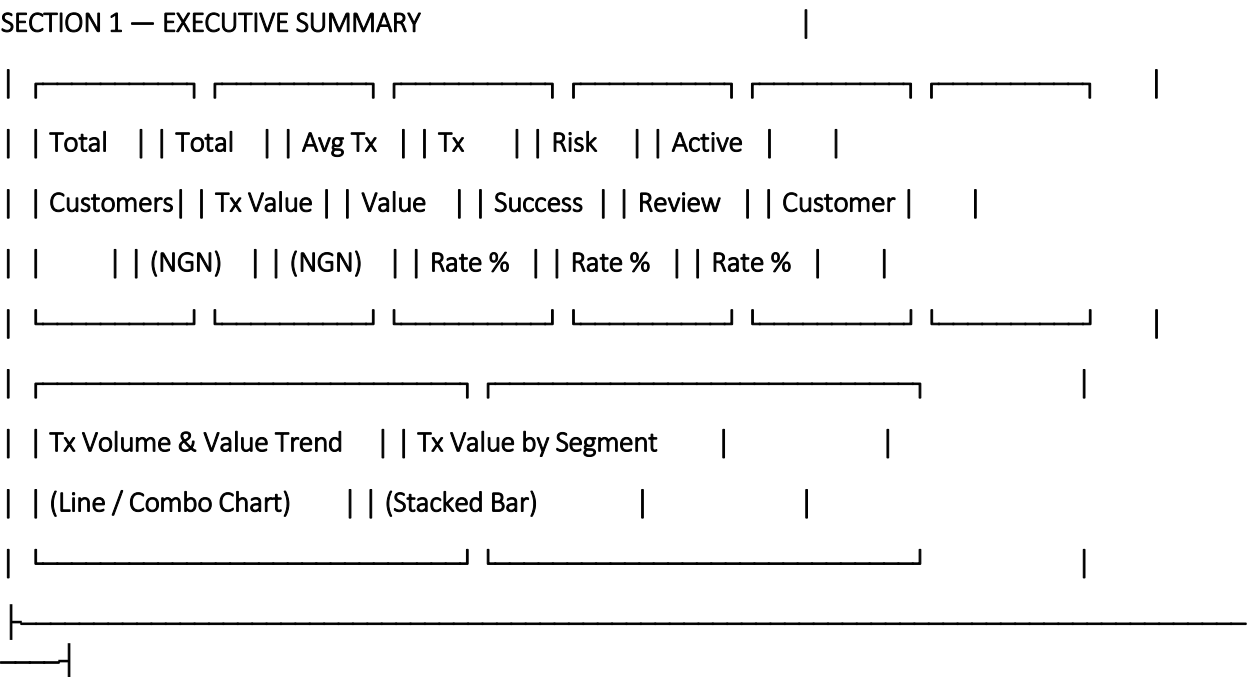
Notes

- Thresholds such as “high-value” and “active customer” should be calibrated after initial exploratory analysis.
- Risk_Review_Flag should be treated as a **review indicator**, not confirmed fraud.
- KPIs should be monitored by segment, channel, device, location, and time period where possible.
- Baselines and targets can be set after Week 2 data profiling.

Part E — Dashboard Planning

Below is a dashboard wireframe for FinTrust's **Business & Data Intelligence Dashboard**. It is designed to be built in Power BI, Tableau, or a similar BI tool and covers customer behaviour, transaction activity/value, channels, status, and risk patterns.

1. Overall Dashboard Layout (Wireframe)



| SECTION 2 — CUSTOMER BEHAVIOUR & SEGMENTATION |

Customers by Segment	Digital Engagement	Tenure vs Activity		
& Account Type	vs Tx Frequency	(Scatter / Heatmap)		
(Donut / Treemap)	(Scatter Plot)			
Customer Activity Rate by Segment, Tenure Band & Account Status				
(Matrix / Heatmap Table)				

| SECTION 3 — TRANSACTION ACTIVITY & VALUE |

Tx Volume by Type & Channel	Avg Tx Value by Segment			
(Clustered Bar)	(Bar Chart)			
High-Value Transaction Share	Tx Value Concentration			
by Segment (Bar)	(Pareto Chart)			

Transaction Volume & Value Over Time by Transaction Type				
(Time Series / Area Chart)				

| SECTION 4 — CHANNEL, DEVICE & STATUS PERFORMANCE |

Transaction Success Rate	Failure/Pending/Reversal			

	by Channel & Device (Bar)	Rate by Channel (Bar)	
	Tx Status Distribution	Avg Tx Value by Channel	
	(Donut)	& Device (Bar)	
	Channel Performance Matrix: Volume, Value, Success %, Risk %		
	(Table with Conditional Formatting)		
	SECTION 5 — RISK & COMPLIANCE		
	Risk Review Rate by	International vs Domestic	
	Segment & Channel (Bar)	Risk Ratio (Bar)	
	Risk Flag by Transaction	Risk Review Trend Over Time	
	Type & Amount Band (Heatmap)	(Line Chart)	
	High-Risk Transaction Table: Tx ID, Customer, Amount, Channel,		
	Device, Location, International, Status, Risk Flag		
	(Drill-through Table)		

2. Dashboard Sections Explained

Section	Purpose	Primary Audience
1. Executive Summary	One-screen overview of customers, transaction value, success, risk, and activity	Executives, Board, Management
2. Customer Behaviour & Segmentation	Understand who customers are and how they engage	Marketing, Customer Insights, Product
3. Transaction Activity & Value	Understand what customers do and where value sits	Finance, Product, Operations
4. Channel, Device & Status Performance	Identify operational friction and channel effectiveness	Operations, Digital, IT
5. Risk & Compliance	Monitor risk review patterns, international exposure, and high-risk transactions	Risk, Compliance, Fraud/Security

3. Key KPI Cards (Executive Summary)

KPI Card	Metric	Visual Type
Total Customers	Count of unique Customer_ID	Card
Total Transaction Value	Sum of Amount_NGN	Card
Average Transaction Value	Amount_NGN ÷ successful transactions	Card
Transaction Success Rate	Successful ÷ total × 100	Card with trend
Risk Review Rate	Risk-flagged ÷ total × 100	Card with trend
Active Customer Rate	Customers with ≥1 tx in period ÷ total × 100	Card with trend

4. Charts and Visuals by Section

Section 1 — Executive Summary

- Line/Combo chart: Transaction volume and value over time.
- Stacked bar: Transaction value by customer segment.

Section 2 — Customer Behaviour & Segmentation

- Donut/Tree map: Customers by segment and account type.
- Scatter plot : Digital Engagement Score vs transaction frequency.
- Heatmap: Tenure band vs activity rate.
- Matrix: Activity rate by segment, tenure, and account status.

Section 3 — Transaction Activity & Value

- Clustered bar: Transaction volume by type and channel.
- Bar chart: Average transaction value by segment.
- Pareto chart: Transaction value concentration.
- Area chart: Volume and value over time by transaction type.
- Bar chart: High-value transaction share by segment.

Section 4 — Channel, Device & Status Performance

- Bar chart: Success rate by channel and device.
- Bar chart: Failure/pending/reversal rate by channel.
- Donut: Transaction status distribution.
- Table: Channel performance matrix with conditional formatting.

Section 5 — Risk & Compliance

- Bar chart: Risk review rate by segment and channel.
- Bar chart: International vs domestic risk ratio.
- Heatmap: Risk flag by transaction type and amount band.
- Line chart: Risk review trend over time.
- Drill-through table: High-risk transaction detail.

5. Filters (Slicers)

Filter	Field	Type
Date Range	Transaction_DateTime	Date slicer
Customer Segment	Customer_Segment	Dropdown / tiles
Account Type	Account_Type	Dropdown
Account Status	Account_Status	Dropdown
Channel	Channel	Dropdown / tiles
Device Type	Device_Type	Dropdown
Location	Location	Dropdown / map
Transaction Type	Transaction_Type	Dropdown
International	International_Transaction	Toggle (Yes/No)
Risk Flag	Risk_Review_Flag	Toggle (Yes/No)
Amount Band	Derived from Amount_NGN	Range slider
Tenure Band	Derived from Tenure_Months	Range slider
Income Band	Monthly_Income_Band	Dropdown

6. Major Insights the Dashboard Should Communicate

1. Customer concentration: Which segments, account types, and cities drive the most customers and transaction value.
2. Engagement reality: Whether Digital_Engagement_Score matches actual transaction behaviour, and where engagement gaps exist.

3. Value hotspots: Which segments, channels, and transaction types generate the highest and most concentrated value.
4. Operational friction: Which channels and devices have the highest failure, pending, or reversal rates.
5. Risk concentration: Where risk review flags cluster — by segment, channel, device, location, amount band, and international status.
6. International exposure: Whether international transactions carry disproportionately higher risk review rates.
7. Customer lifecycle: How tenure and account status relate to activity, value, and risk.
8. Actionable priorities: Which customers, channels, or transaction patterns need immediate attention from risk, operations, or customer support teams.

7. Design Principles Applied

- Drill-down: Executive summary → segment → transaction detail.
- Colour coding: Green = healthy, Amber = watch, Red = risk/failure.
- Consistency: Same KPIs and filters across all sections.
- Audience layering: Executives see Section 1; analysts drill into Sections 2–5.
- Data governance: PII masked in tables; role-based access for risk and compliance views.
- Refresh cadence: Daily for transaction/risk views; weekly for customer segmentation.

This dashboard would turn FinTrust's customer and transaction data into a single, decision-ready view for management, risk, operations, and customer support team

